Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

Information leaflet for Parent/Guardian

Study title: Investigating the effect of spatially complex auditory environments on neural speech processing: a study on neurodivergent individuals.

Principal investigator's names: Giovanni Di Liberto
Principal investigator's titles: Assistant Professor

Contact details of principal investigator:

Co-investigator's names: (1) Amirhossein Chalehchaleh
(2) Sharon Moran

Co-investigator's titles: (1) Postdoctoral Researcher, Trinity College Dublin
(2) Research assistant, Trinity College Dublin

Introduction

Understanding speech in everyday noisy environments is an important part of communication, learning, and social interaction. While many people can focus on a conversation despite background noise, this can be much more challenging for neurodivergent individuals. Improving our understanding of how the brain processes speech in these complex listening situations may help guide future approaches and technologies to support communication in school, at home, and in the community.

This study aims to investigate how neurodivergent children (ages 7-18 years) process and understand speech across different listening conditions. We will compare listening performance across different sound environments (loudness, sound locations) to better understand the factors that make speech easier or more difficult to understand. Participants will complete a series of short listening activities in a child-friendly research setting. They will listen to brief spoken stories presented under different listening conditions and answer simple questions about what they heard. Brain activity will be recorded using non-invasive electroencephalography (EEG) during the listening tasks.

To provide comparison data, we also require adult participants.Taking part in this study is entirely voluntary, and parental or guardian consent is required before a child can participate.

What is the purpose of this study?

Many children find it difficult to understand speech in places such as classrooms, playgrounds and busy public spaces. This research aims to understand the brain and behavioural processes involved in listening so that future interventions and assistive technologies can be better designed.

Who can take part in this study?

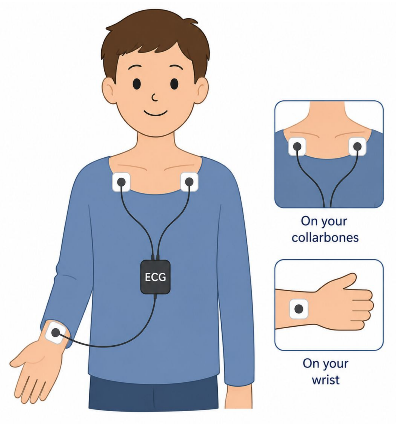
We are looking for native English-speaking neurodivergent children and young people aged 7–18 years, as well as native English-speaking adults.

What will happen in the study?

The study will be held at the Lloyd Institute, Trinity College Dublin for a duration of approximately 1 hour in total.

<u>ECG Data Collection</u>

- Three small ECG sensors will be fitted using adhesive stickers to record the participant's heart activity during the study: two near the collarbones and one at the wrist.



Example of ECG sensors worn in the study.

<u>Listening task</u>

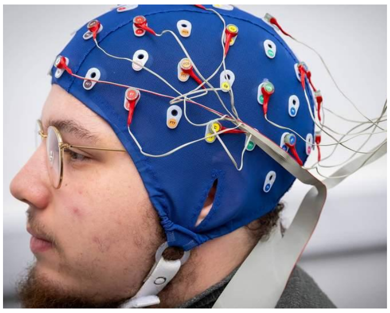
- The participant will complete a brief sound-location task, played through speakers. Headphones will also be worn during the task.
- Participants will listen to short spoken stories under different listening conditions that vary in background noise, loudness and the perceived location of sounds.
- After each story they will answer simple questions about what they heard.
- This listening task will last approximately 30 minutes, with a further 30 minutes for set up and final questions.

*** Optional second visit at a later date**

We will ask participants if they would like to return a second time to do the experiment with and EEG cap to record the brain's electrical activity during the listening tasks.

<u>EEG Data Collection (Later experiment, optional**)</u>

- The researcher will gently fit an EEG cap onto your child's head. A small amount of water-soluble gel will be applied through the sensor holes so they make good contact with the scalp to record brain activity. After the listening task, shampoo, towels and hair-drying facilities will be available if you wish to wash the gel from your/your child's hair.



Example of EEG cap worn in the study.

What are the benefits of this study?

Each participant All4One voucher: €20 for adults 18+ / €30 for children aged 7-18 years.

Are there any risks to taking part?

The study is non-invasive. Participants may take breaks or stop the study at any time.

Is the study confidential?

All data collected for this study will be treated as confidential according to the Trinity College Dublin policies. Confidentiality may be breached in circumstances required by law where information emerges that raises a concern for the safety or well-being of any individual. The information we collect about the participants (brain activity data, behavioural responses to the tasks and study-relevant demographics data) will be held on a secure database in Trinity College Dublin School of Computer Science and Statistics. The data will be coded with a unique ID and held on password-protected computers that are only accessible to the study researchers. The key to this coding will be kept on an encrypted Excel file on the secure OneDrive system in Trinity College Dublin, and will be accessible by members of the research team only. This key will be destroyed when the study is finished. Online data will be encrypted at rest and in transit to ensure security and privacy. Only the researchers on this project and their collaborators will have access to the data from this study. If you consent to being in this study, it is important that you are aware of the procedure in the case of any incidental findings.

What results will I get?

We are not planning to routinely provide feedback on the results of the computer testing. This is because the individual results on their own cannot give us much information, and we will be looking at all of the results of all of the participants together (i.e. we look at the response of a group). We are hoping to publish the results of the study in scientific journals and to present the results at scientific conferences. There will be no information capable of identifying the young person in any publications or presentations.

What do I do if I no longer want to be involved in research?

It is possible to leave the study at any time without giving any reason, and without penalty. To leave the study, the parent / guardian may contact the investigators, (contact details below). Even after participating, you can withdraw your data up to the point that the results from different people are added together. It is important to know that declining or withdrawing participation will not impact on any service provision.

Data Protection Information

Information collected for this study, including EEG recordings, behavioural responses and relevant demographic information, will be processed for scientific research purposes, stored securely within Trinity College Dublin, and retained in accordance with university policy. Personal information will be protected using coded identifiers and handled in line with data protection legislation.

Your rights in relation to Data Protection

- Withdraw consent at any time.
- Request access to your data.
- Request correction of inaccurate information.
- Request deletion or restriction of personal data where applicable.
- Ability to lodge a complaint with the Data Protection Commission.

How long do you keep the data collected for this study?

The EEG and behavioural data will be stored in a secure database at Trinity College Dublin (TCD) for ten years to support ongoing scientific research. With your consent, the data collected during this study may also be used in future ethically approved research related to speech processing. Once the study has been completed, the data will be anonymised, meaning it will no longer be possible to link the data to your personal details.

Who is funding the study?

This study is funded through the ARC Hub ReFocus programme for hearing research.

Complaints and further information

If you have questions or wish to make a complaint, please contact Dr Giovanni Di Liberto or a member of the research team; Postdoctoral Researcher Amirhossein Chalehchaleh: achalehc@tcd.ie
Or Research Assistant Sharon Moran: moransj@tcd.ie